UNJAM: THE LAST SLOT — UI / UX / Design Specification (current state)

*Version of the game described: 0.1.0 (release candidate) · Godot 4.7.1 · **revision 3 (2026-10-17): after the second review pass** — amber triangle fill/rim, stronger tray outline, hint-state and dialog verification tests; revision 2 added contrast tokens, tray rule tags, status pills, action-bar order, reward animation. Screens in `docs/design/` were re-rendered from the updated game. Purpose: hand this file (plus the images in `docs/design/`) to a design reviewer — human or AI — and ask what should change. Everything here describes **what exists today**, with exact values taken from the source; nothing is aspirational. Section 13 lists weaknesses I measured or noticed, and Section 14 is a ready-to-paste review request.*

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0. What is in the design pack

File	What it is
`docs/UI_UX_DESIGN_SPEC.md`	this document
`docs/design/sheets/sheet_1_menus_levelselect_tutorial.png`	contact sheet: splash, menus, settings, level select, credits, tutorial intro (screens 01–10)
`docs/design/sheets/sheet_2_gameplay_and_hints.png`	contact sheet: gameplay states, restart/hint dialogs (screens 11–20)
`docs/design/sheets/sheet_3_mechanics_dialogs_results.png`	contact sheet: mechanics levels, pause, rules, win, fail (screens 21–29)
`docs/design/sheets/sheet_4_amber_triangle_before_after.png`	all 8 shapes, and the amber triangle in every state (normal, selected, locked, frozen, on page background): **top = before, bottom = after**
`docs/design/screens/NN_*.png`	the 29 individual screens, 1080×1920 (= the 720×1280 design canvas ×1.5)
`docs/design/UNJAM_UI_UX_Design_Spec.pdf`	**single self-contained PDF (21 pages): this spec + all 29 screens embedded — easiest to hand to a reviewer**
`docs/design/UNJAM_design_pack.zip`	all of the above in one file (not tracked in git; rebuild by zipping the spec + `docs/design/{sheets,screens}`)

All screenshots are rendered by the **real game screens** on a desktop OpenGL context (not photographed on a phone), so fonts/antialiasing may differ slightly from a device. Regenerate with `tools/make_design_screens.gd` (instructions in §15).

1. Product context (for the reviewer)

- **Genre:** casual sorting / tray-capacity puzzle for Android phones, portrait, one-handed thumb play, offline, no ads, no account, no permissions.
- **Core idea:** move the top object of a tray onto the same shape or into an empty tray; fill a tray with one shape to clear it; clear everything to win. Trays have limited slots; the tension is **"who gets the last free slot?"** (hence the HUD chip *Free slots → Last slot!*).
- **Content:** 40 levels in 7 stages (Basics → Tight Trays → Move Limits → Locks → Special Trays → Ice → Mastery). Each new mechanic is introduced once by a card.
- **Meta:** stars per level (1–3), coins (earned on first clear, spent on hints), unlock-next progression, local save.
- **Audience intent:** general casual audience, calm tone, premium-feeling but minimal. No violence, text-light, shape-based.
- **Tone of voice:** short, friendly, plain sentences ("That tray is full.", "Let's go", "Not now"). No exclamation spam; one "clear!" on win.

2. Platform and display facts

Property	Value
Engine / renderer	Godot 4.7.1, GL Compatibility (2D only)
Design canvas	**720 × 1280 logical px** (9:16). Stretch mode `canvas_items`, aspect `expand`
Taller phones (e.g. 20:9)	the canvas grows vertically (720×1560…); content is laid out with containers, the board is centred in the free space
Wider windows (tablet / foldable / landscape — Android 16 ignores the portrait lock ≥ 600 dp)	content is held in a **centred column** (max 780 logical px wide; 900 on level select) that re-flows on resize/rotation; side areas show the plain background
Orientation	portrait requested (`screenOrientation=portrait`)
Safe area	system-bar/cutout insets are added to the top/bottom (and sides) margins (`ColumnMargin`); the maths is unit-tested, **real device values not yet verified**
Min touch target	**96 logical px** (= 48 dp on a 360 dp-wide phone). Trays are tapped on their whole column cell (much larger)
Input	single tap only (`InputEventMouseButton`, Android touch is emulated as mouse). No drag, long-press, swipe or multi-touch gestures; no keyboard
Back button	closes a dismissible dialog → in-game opens Pause → Level select/Credits go back to Menu → Menu exits. Result dialogs (win/fail) ignore it

Property	Value
Scroll	Level select and Credits scroll vertically by touch-drag (dead-zone 24 px so a wobbly tap is still a tap)
Fonts	Godot’s built-in default UI font (Open Sans family) — no custom font files. Bold = synthetic embolden (0.55) applied to text ≥ 34 px and to all buttons

3. Design principles / visual identity

1. **Calm paper-and-ink look:** warm neutral background, white cards, one coral accent, deep indigo text.
2. **Shapes before colours:** every object type has a unique **shape + colour** so the game is readable without colour vision.
3. **Tactile 2D:** rounded rectangles (radius 26–40), soft drop shadows under buttons/cards, a small specular dot and darker rim on objects, soft ellipse shadow under objects.
4. **Everything procedural:** no bitmap art in the UI — all shapes, icons, the logo and the coin/star glyphs are drawn in code (crisp at any size); audio is synthesised in code.
5. **Restraint:** at most one primary (coral) button per screen/dialog; the HUD shows only level, moves, free slots, coins and three actions.
6. **Logo idea:** a tray holding three objects (circle, square, triangle) and one **empty dashed ring** — “the last slot”.

4. Design tokens

4.1 Colour (exact values from `src/ui/ui_theme.gd`)

Token	Hex	Usage
`BG`	`#F3F0EA`	screen background (warm off-white)
`BG_DARK`	`#E7E2D8`	pressed state of secondary/icon buttons
`PANEL`	`#FFFFFF`	cards, dialogs, chips, secondary buttons, level tiles
`TRAY`	`#E1DCCF`	tray body, progress-bar track
`TRAY_OUTLINE`	`#857D6C`	tray boundary, 4 px (3.59:1 on BG)
`TRAY_EDGE`	`#9C9482`	chip/button outlines (2.65:1 on BG); slot guide rings use it at 42 %
`INK`	`#2B2F4A`	primary text, icons, dashed “last slot” ring, toast background, padlock
`INK_SOFT`	`#5A5E7A`	secondary text (level name, captions, stage headers, stats) — 5.57:1 on BG
`ACCENT`	`#D2432F`	primary button fill (white text 4.58:1), selection ring, current-level tile border, progress fill

Token	Hex	Usage
`ACCENT_DARK`	`#A82F21`	primary button pressed, “THE LAST SLOT” tagline (5.97:1), warning text
`GOOD`	`#2FBF71`	valid-destination highlight and tray-clear flash only (never behind text)
`GOOD_DARK`	`#178348`	switch ON track (white label 4.80:1)
`SWITCH_OFF`	`#756E5D`	switch OFF track (white label 4.9:1)
`WARN`	`#F4B400`	hint highlight border, coin, stars, hint cost badge
`ICE`	`#BFE6FA`	frozen overlay (with `#7CC3E8` rim and `#4AA3D6` snowflake)
Danger button	`DANGER #C23A3A` / pressed `#A02E2E`	destructive confirmations (“Restart”, “Erase everything”); white text 5.30:1
Chip alert fills	`#FFD9D3` (≤ 3 moves left) · `#FFE9A8` (last slot)	HUD chips
Dim behind dialogs	`#1A1C33` @ 55 %	modal scrim
Locked tile	fill `#DCD6C7`, number `#7D7863`	locked level tiles (number 3.06:1)

4.2 Object palette (8 types: colour + shape; each has a rim 36 % darker and a light sheen — only the amber triangle gets a 50 % darker, thicker rim, tokens `RIM_DARKEN` / `RIM_INSET`)

Type	Colour	Shape
0	`#EF5B5B` red	circle
1	`#2E9BE6` blue	rounded square
2	`#EAA600` amber (was `#F5B400`)	triangle — rim 50 % darker and thicker
3	`#2FBF71` green	hexagon
4	`#8E5CF5` purple	diamond
5	`#F06BB5` pink	star (5-point)
6	`#10A9A0` teal	pentagon
7	`#F2792B` orange	plus/cross

Levels use 2–5 types (level 1 uses 2; most later levels use 4–5), trays have capacity 3, 4 or 5, and a level has 4–7 trays.

4.3 Typography (logical px at the 720 canvas)

Role	Size	Weight	Used for
Logo wordmark “UNJAM”	128	bold	menu/splash
Tagline “THE LAST SLOT”	38	bold, `ACCENT_DARK`	menu/splash

Role	Size	Weight	Used for
H1	44	bold	screen/dialog titles, “Level N”, Play button text
H2	34	bold	buttons, coin balance, “+N coins”
Body	28	regular	dialog text, chips, toasts, coach text, stage headers
Small	22 (24–26 for dense text)	regular	level name, captions under icons, rules text, credits
Micro	16–20	regular	switch “ON/OFF”, hint-cost badge

Line wrapping: word-smart; text is centred in dialogs, left-aligned in Credits and Level-select stage headers. No dynamic font scaling.

4.4 Shape, spacing, elevation

Element	Radius	Border	Shadow
Buttons (primary/danger/secondary)	30	secondary: 2 px `TRAY_EDGE`	soft: 10 px blur, +5 px y, ink @ 18 % (ghost buttons: none)
Dialog card	40	none	soft (same)
Tray	0.34 × slot size	4 px `TRAY_OUTLINE`	none
Chips (moves/free slots/coins)	28–30	2 px `TRAY_EDGE`	none
Toast	28	none	none
Level tile	30	2 px `TRAY_EDGE` (current: 5 px `ACCENT`)	soft when unlocked
Switch track	26	none	none

Spacing (logical px): screen margins 20 (game) / 24–40 (other screens) plus safe-area insets; VBox gaps 12–18; dialog padding 36 horizontal × 34 vertical, 18 between blocks, 14 between buttons; HUD button gap 18; level-grid gap 14.

4.5 Iconography (all drawn in code, 3–4 px strokes, rounded caps)

Pause (two bars) · Back (chevron) · Undo (counter-clockwise arrow) · Restart (clockwise arrow) · Hint (light bulb) · Coin (gold disc with inner ring) · Star (filled gold / empty `#D8D2C4`) · Padlock (with the **key shape inside**) · Snowflake (ice) · Sound/Music (+ crossed variants, currently unused in UI).

4.6 Motion (all durations in seconds; Reduce motion setting makes everything instant, no shake, no pulse, no screen fade)

Event	Animation
Select object	lifts 0.45 × slot above its slot, 0.10 s ease-out; white disc + coral ring behind it
Move	3 stages: rise to above the source tray (0.09) → glide above the destination (0.13, ease-in-out) → drop with a small bounce (0.10) ≈ 0.32 s total; **all input locked meanwhile**

Event	Animation
Tray complete	objects pop to 1.35× and fade (0.16 / 0.20, delayed 0.12) while the tray flashes green (0.40)
Invalid move	tray (and its top object) shakes ±8 → ±5 px, 5 × 0.04 s, + toast
Board (re)build after undo/restart	objects scale 0.4→1 with back-out + fade-in (0.18 / 0.14)
Hint	source object pulses 1.0↔1.15 (0.4 s each, looping) until the next tap; both trays get a yellow outline
Dialog open	card scales 0.92→1 (back-out 0.16) with fade 0.12
Screen change	cross-fade through the background colour, 0.14 s out + 0.14 s in
Toast	appears instantly, holds 1.2 s + 0.05 s per character (min 1.9 s, max 4.5 s), fades 0.3 s

4.7 Audio (all synthesised in code; SFX at −6 dB, music at −16 dB; toggles for each)

Cue	Sound design
tap (buttons)	very short blip, 660→560 Hz, 50 ms
select	rising soft tone 520→780 Hz, 90 ms
move	soft falling “thud” 400→300 Hz + 300→260 Hz, ~160 ms
invalid	low soft buzz 200→150 Hz, 140 ms
undo	falling tone 500→330 Hz, 110 ms
clear (tray)	three-note rising arpeggio C5-E5-G5
coin	two quick high notes 988 / 1319 Hz
win	C5-E5-G5-C6 fanfare
fail	descending 330 → 262 → 196 Hz
music	8-second loop, A-minor / F / C / G arpeggio (sine), very quiet

5. Component specifications

Component	Spec
Primary button	coral `ACCENT`, white bold 34 px text (44 px for the main *Play/Continue*), min height 96 (112 for Play), radius 30, side padding 28, soft shadow; pressed `ACCENT_DARK`; disabled 35 % alpha
Secondary button	white, ink bold text, 2 px `TRAY_EDGE` outline, soft shadow; pressed `BG_DARK`
Ghost button	transparent, ink bold text, no shadow; pressed 6 % black — used for tertiary actions (“Settings”, “Privacy & credits”, “Level select”, “Reset all progress”, “Close”)

Component	Spec
Danger button	red `#D94A4A`, white text — only on irreversible confirms
Icon button (HUD)	150 × 112 white card, radius 30, 2 px outline + shadow; vector icon centred at 38 % height, 22 px caption at 86 %; disabled = 35 % alpha content; optional corner **badge** (hint cost “15” on a gold pill)
Square icon button	96 × 96 (pause, back)
Chip	pill 250 × 56 min, white, 2 px outline, 28 px ink text; alert fills coral-tint (≤ 3 moves left) or yellow-tint (*Last slot!*)
Coin chip	pill ≥150 × 96: gold coin icon + balance (34 px bold)
Switch row	96-high rounded row on `BG`, label 28 px left, track 96 × 52: ON green with right knob and “ON”, OFF grey `#C9C3B4` with left knob and “OFF” — whole row is the touch target
Dialog (Modal)	scrim + centred white card, min width 600, radius 40, title 44 px bold, optional body (28 px, centred), then a stack of full-width buttons (primary on top, then secondary, then ghost). Tapping the scrim dismisses dismissible dialogs; result dialogs (win/fail) are not dismissible
Toast	dark-ink pill, white 28 px text, bottom-anchored just above the action bar (game) / list bottom (level select)
Level tile	150 × 146, number 44 px, three small stars under it (gold = earned); **current** level has a 5 px coral border; **locked** tile is a darker beige with a padlock instead of stars and a grey number
Tray	vertical rounded container: width = 1.3 × slot, height = capacity × slot + 0.3 × slot; one faint **slot ring** per capacity slot (a non-colour capacity cue); objects stack bottom-up; tap area = the full column cell
Object	circle/shape of radius 0.40 × slot with rim, sheen dot and ground shadow, always drawn at ≥ 72 % opacity so its shape stays recognisable; **locked** = 72 % opacity + a white **status pill** under it (padlock glyph + the *shape that opens it*); **frozen** = pale-blue ring + a pale-blue status pill (snowflake + moves left); **selected** = white disc + 5-px coral ring

Component	Spec
Tray rule tag	white 0.31 × slot badge with ink ring straddling the tray's top edge: shows the single **shape** a reserved tray accepts, or an **up-arrow** for a one-way tray. Replaces the earlier faint ghost shape / tiny triangle
Slot size	computed to fit: `clamp(min(height-fit, width-fit), 36, 112)` logical px (row gap 40 px so tags never touch the tray above; the board sits at 58 % of the free height, slightly low, nearer the thumb); the tray grid is 1 row for ≤ 4 trays (level 1), 3 columns × 2 rows for 5–6 trays, 4 columns × 2 rows for 7 trays (the largest shipped level)

6. Information architecture and navigation

```
Splash ──(≈1.4 s auto)──► Main menu ──► Play/Continue ──► Game (level N)
                          │   ├──► Levels ──► Game (any unlocked level)
                          │   ├──► Settings (dialog) ─► How to play (dialog) / Reset progress
(confirm)
                          │   └──► Privacy & credits
Game ─► Pause (dialog: sound, music, reduce motion, Resume, How to play, Level select)
     ─► Restart (confirm if moves > 0) · Undo · Hint (confirm) · result dialogs (Win: Next / Play
again / Level select · Fail: Undo / Restart / Level select)
```

Depth is never more than 3 taps from any state; there is no bottom tab bar and no account/profile area.

7. Screen-by-screen specification

(Image names refer to `docs/design/screens/`.)

7.1 Splash — `01_splash.png`

Background `BG`; the logo (tray with circle/square/triangle + dashed ring, "UNJAM", "THE LAST SLOT") centred and fading in over 0.35 s; small grey "Loading…" near the bottom. Holds ≈ 0.5 s after the fade, then goes to the menu. If the level file cannot load, the text changes to an error message in `ACCENT_DARK` (never expected in release).

7.2 Main menu — `02_menu_new_player.png`, `03_menu_with_progress.png`

Top-right: coin chip. Then: logo (≈470 px tall), progress text ("14 / 40 levels cleared · 38 stars", 24 px soft), progress bar (18 px, coral fill on tray track), **primary button** — *Play* for a new player or *Continue · Level N* (112 px tall, 44 px text), secondary *Levels*, then two ghost buttons side by side: *Settings*, *Privacy & credits*. The lower third of tall phones is empty background. If the save file was damaged and a backup restored (or reset), a one-line warning in red appears under the buttons.

7.3 Settings dialog — `04_settings_dialog.png`

Title *Settings*; three switch rows (Sound effects · Music · Reduce motion); secondary *How to play* (opens the rules dialog); ghost *Reset all progress* (opens a confirmation); primary *Done*.

7.4 Reset confirmation — `05_reset_progress_confirm.png`

Reset progress? — "This erases completed levels, coins and settings on this device. It can't be undone." Danger button *Erase everything*, secondary *Cancel*.

7.5 Level select — `06_level_select_top.png`, `07_level_select_locked_toast.png`

Header: square back button · "Levels" · total stars "★ 38". Scrolling list grouped by stage: grey header "Stage 4 · Locks" then a 4-column grid of tiles (see §5). Opens scrolled so the current level is in view. Tapping a locked tile plays the invalid sound and shows the toast "Clear level N to unlock this one." where N is the highest level currently unlocked (the one still to be cleared); the tile itself does not move. Completed levels show their stars; there is no separate "completed" tick.

7.6 Privacy & credits — `08_privacy_and_credits.png`

Back button + title; one white card with left-aligned 24 px text in four blocks: PRIVACY, CREDITS, OPEN SOURCE (Godot MIT notice), VERSION. Scrolls.

7.7 Game HUD (all gameplay screens) — see `10_…` onward

Top bar (left→right): **Pause** (96 sq) · centre title column ("Level 15" 44 px bold over the level name 22 px soft) · **coin chip**. Below it two centred chips: **Moves: N** (or **Moves left: N** on limited levels; turns coral-tinted at ≤ 3) and **Free slots: N** (reads **Last slot!** with a yellow tint when exactly 1). Optional **coach line** (28 px, centred) only during the level-1 tutorial. The **board** fills the remaining space (see §5/§8). **Bottom bar: Restart, Undo, Hint (15)** icon buttons (Restart is deliberately far from Hint). A toast appears just above the bottom bar.

7.8 Tutorial — `09_intro_card_how_to_play.png`, `10_game_level1_initial_with_coach.png`, `11_…`

On the very first level: a **How to play** card ("1. Tap a tray to lift its top object. 2. Tap another tray to drop it there. …") with *Let's go*. Then a **coach line** under the chips that changes with state: "Tap the glowing tray to lift its top object." → (after lifting) "Now tap the highlighted tray to drop it there." The source tray glows yellow, the target tray gets a yellow outline. The coach disappears after level 1 is cleared. Later mechanics get a one-time **intro card** on their first level (§10): `19_level11_move_limit_intro.png`.

7.9 Gameplay states

State	Image	What the player sees
Idle	`10_…`	all objects on slots; Undo disabled until a move exists
Selected	`11_…`	lifted object with coral ring; **valid** destination trays get a green outline + light green fill; invalid trays unchanged
Invalid drop	`12_…`	tray shakes, toast with the reason, selection stays

State	Image	What the player sees
Moving	`13_…`	object in flight; all buttons and trays inert; destination highlights already cleared
After move	`14_…`	counters updated, Undo enabled
Tray cleared	`27_…`	objects pop, tray flashes green, objects removed
Restart confirm	`15_…`	only when ≥ 1 move has been made

7.10 Hint flow — `16`, `17`, `18`

Tap **Hint** → the game computes a real winning move silently → dialog *Need a hint?* "Show a good next move for 15 coins? You have 85." (*Show hint (15)* / *Not now*) → source object pulses and both trays get a yellow outline until the next tap; coins −15. If coins < 15: *Not enough coins* — "A hint costs 15 coins and you have N. Complete levels to earn more." (*OK*). If the position can never be finished: *This can't be finished* + explanation (free) with *Undo last move* / *Restart level* / *Close*. If no hint exists: *No hint right now* (free).

7.11 Mechanic levels — `20_level15_padlocks`, `21_level21_reserved_trays`, `22_level24_one_way_tray`, `23_level29_ice_frozen_objects`, `24_level40_dense_mastery`

- **Padlock:** the locked object stays visible (72 % opacity) with a white pill beneath it: padlock glyph + the **shape that opens it**; clearing a tray of that shape removes the lock.
- **Reserved tray:** a white tag badge on the tray's top edge showing the only shape it accepts.
- **One-way tray:** a slightly darker tray with an up-arrow tag on its top edge; objects can be taken but never dropped in.
- **Ice:** pale-blue ring around the object and a pale-blue pill (snowflake + the number of moves left until it thaws); the shape underneath stays fully visible.
- **Move limit:** chip switches to "Moves left: N".

7.12 Pause — `25_pause_menu.png`

Paused: three switches, primary *Resume*, secondary *How to play*, ghost *Level select*. Also opened by the Android back button during play.

7.13 How to play (rules) — `26_how_to_play_rules_dialog.png`

Verbatim rules text (see §10.3) with *Got it*.

7.14 Level complete — `28_level_complete_dialog.png`

Non-dismissible dialog: "Level N clear!", three big stars (earned = gold, others grey; middle one larger), "Solved in 8 moves (par 8)", "+30 coins" (first completion) or "Replay: no new coins", buttons *Next level* (primary; replaced by a "You finished every level" line on level 40), *Play again*, *Level select*. Stars pop in one after another (0.28 s each, 0.2 s apart) and "+N coins" counts up over 0.55 s while the coin sound plays; with **Reduce motion** the final values show immediately.

7.15 Level failed — `29_level_failed_dialog.png`

Non-dismissible dialog: "Out of moves" (move limit) or "No moves left" (deadlock) with one explanatory sentence, *Undo last move* (primary; shown only if undo is possible), *Restart level*, *Level select*.

8. Interaction model

Two-tap move: tap tray A (lifts its top object) → tap tray B (drops it). Tap A again to cancel. There is no drag-and-drop.

Situation	Result
Tap an empty tray as source	shake + toast “That tray is empty.” (reason `EMPTY_SOURCE`, §10.4)
Tap a tray whose top object is locked/frozen	shake + toast explaining lock/ice
Tap a destination that refuses the object	shake on that tray + toast with the reason; the source stays selected
Tap during an animation / while a dialog is open	ignored
Undo	free, unlimited (up to 400 moves), restores moves, trays, unlocked state, ice countdown; allowed after a failure
Restart	confirm dialog if moves > 0, else immediate
Win	evaluated only from game state; reward processed exactly once
Fail	out of moves or no legal move; Undo recovers

State machine: `Idle ⇄ Selected → Animating → (Idle | WinDialog | FailDialog)`; dialogs sit on top of Idle. `Animating` and any open dialog lock all game input and disable Hint/Undo/Restart.

Feedback matrix | Event | Visual | Audio | Text | |——-|——–|——-|——| | Select | lift + ring + green targets | select | coach (tutorial only) | | Valid move | 3-stage glide + bounce | move | counters update | | Tray cleared | pop + green flash | clear | — | | Invalid move | shake | invalid | toast with reason | | Undo | board rebuilds with pop-in | undo | — | | Hint paid | pulse + yellow outlines | coin | — | | Win | dialog | win (+coin) | stars, moves, coins | | Fail | dialog | fail | explanation | | Button press | pressed colour | tap | — | | No haptics anywhere | | | |

9. Progress, economy and number display

- Start balance 30 coins; hint 15; first clear pays 15 + 5 × stars (20–30); replays pay 0 (they can only improve stars).
- Stars: ≤ 125 % of par → 3; ≤ 175 % → 2; otherwise 1 (par = proven shortest solution length). Level select shows best stars.
- Move limits (23 of the 40 levels have them: 11–14, 17–20, 23, 25, 26, 29–40) are par + max(spec slack, 45 % of par). Undo refunds moves.
- Level unlock: clearing level N unlocks N+1; locked tiles are visibly different; “Continue” opens the current/next unlocked level.

10. Copy deck (verbatim, generated from the code)

10.1 Stage names

Stage	Name
1	Basics
2	Tight Trays
3	Move Limits
4	Locks
5	Special Trays
6	Ice
7	Mastery

10.2 One-time intro cards (`CampaignText.INTROS`)

Key	Title	Body
`accepts`	Reserved trays	A tray with a shape tag on its top edge only accepts that shape. Use it to park the right objects.
`basics`	How to play	1. Tap a tray to lift its top object. 2. Tap another tray to drop it there. Objects can land on the same shape, or in an empty tray. Fill a tray with one shape to clear it. Clear every tray to win.
`capacity`	Mind the free slots	Every tray has a limited number of slots. Watch the Free slots counter: when only the last slot is left, choose carefully what goes in it. Made a mistake? Undo is free.
`ice`	Ice	A frozen object can’t move until enough moves have been made. The number shows how many moves remain before it thaws.
`limit`	Move limits	Some levels allow only a set number of moves. The counter shows how many you have left. Undo gives moves back, so planning ahead is always allowed.
`lock`	Padlocks	A padlock freezes an object in place. The shape on the padlock tells you what opens it: clear a tray of that shape and the lock disappears.

Key	Title	Body
`mix`	Everything together	From here on, levels mix locks, ice, reserved and one-way trays. Plan the last slot before you fill it.
`source`	One-way trays	A tray with an up-arrow tag only gives objects away. You can take from it but never drop into it.

10.3 Rules text (“How to play”)

> Tap a tray to lift its top object, then tap another tray to place it.
>
> • Only the top object of a tray can move. • An object can land on the same shape, or in an empty tray. • Trays have limited slots. A full tray can’t take more. • Fill a tray with one shape and it clears. • Clear every tray to win.
>
> Special rules: • Padlock: clear a tray of the shown shape to open it. • Ice: wait for the shown number of moves. • Shape tag on a tray: it takes only that shape. • Up-arrow tag on a tray: you can only take from it. • Move limit: finish before the counter reaches 0.

10.4 Refusal reasons shown as toasts

Reason code	Text
`BAD_TRAY`	That isn’t a tray.
`EMPTY_SOURCE`	That tray is empty.
`FROZEN`	Frozen! It thaws after the number of moves shown.
`FULL`	That tray is full.
`GAME_OVER`	This level is already finished.
`LOCKED`	Locked! Clear a tray of the shape shown on the padlock to open it.
`NO_MOVES_LEFT`	No moves left.
`SAME_TRAY`	*(silent — tapping the selected tray just deselects)*
`SOURCE_ONLY`	This tray only gives objects away (up-arrow tag). You can’t drop into it.
`TYPE_MISMATCH`	Objects only stack on the same shape, or in an empty tray.
`WRONG_TYPE_FOR_TRAY`	This tray only takes the shape shown on its tag.

10.5 Other fixed strings

Menu: *Play* · *Continue* · *Level N* · *Levels* · *Settings* · *Privacy & credits* · "N / 40 levels cleared · N stars". HUD: *Level N* · *Moves: N* · *Moves left: N* · *Free slots: N* · *Last slot!* · *Undo* · *Hint* · *Restart*. Dialogs: *Restart level?* "Your progress in this level will be lost." (*Restart* / *Keep playing*) · *Need a hint?* · *Not enough coins* · *This can't be finished* · *No hint right now* · *Paused* · *Settings* · *Reset progress?* · *Level N clear!* · *Out of moves* / *No moves left* · *Next level* · *Play again* · *Level select* · *Undo last move* · *Restart level* · *Let's go* · *Got it* · *OK* · *Done*. Toasts: "Clear level N to unlock this one." (level select) · "That hint is no longer valid. You weren't charged." · "Not enough coins." Tutorial coach: "Tap the glowing tray to lift its top object." · "Now tap the highlighted tray to drop it there." · "Tap the lifted tray again to put it back, then try the glowing one." Credits/privacy text is in `src/ui/credits_screen.gd`.

11. Accessibility status

Provided: unique shape per object type (colour is never the only cue) · capacity drawn as slot rings · locks show their key shape · ice shows a number · "Last slot!" has text, not just a colour · Reduce-motion setting · 48 dp touch targets · no time pressure, no hidden gestures, no flashing > 3 Hz · sound and music optional and the game is fully playable muted · text sizes 22–44 px on a 720 px canvas (≈ 11–22 sp on a 360 dp phone).

Measured contrast (WCAG 2.x ratio; thresholds: 4.5 normal text, 3.0 large text/graphics; enforced by `tests/test_contrast.gd`):

Pair	Before	Now	Verdict
Ink text on background / on white	11.48 / 13.06	11.48 / 13.06	pass
Soft grey text on background / white	4.33 / 4.92	**5.57 / 6.33**	pass (was failing on BG)
White on primary coral button	2.80	**4.58**	pass (was failing)
White on pressed coral	3.87	**6.79**	pass
White on danger button	4.18	**5.30**	pass
Tagline on background	3.40	**5.97**	pass
"ON"/"OFF" label on switch	2.38	**4.80 / 5.07**	pass
Coral graphics (ring, current tile, progress) on background	2.5	**4.03**	pass (≥ 3)
Tray outline on background	1.47	**3.59**	pass (3:1 graphical boundary) — tray *fill* vs page is 1.20:1 but the boundary is the outline
Locked-tile number on locked tile	2.09	3.06	disabled state, now ≈3
Object rims vs tray fill	amber fill 1.34	rims **3.90–6.00** (amber rim 5.12); amber fill 1.54	every shape's boundary ≥ 3:1 (enforced); amber fill itself stays low, so it is carried by its dark rim and unique triangle shape

Pair	Before	Now	Verdict
Ink on alert chips	10.0 / 10.9	same	pass

Not provided: screen-reader/TalkBack support (the engine version used has no Android accessibility server), system font-size scaling, high-contrast mode, haptics, dyslexia-friendly font option, left-handed layout variant (the bottom bar is centred).

12. Responsive behaviour (renders inspected)

Window	Logical canvas	Result
720 × 1280 (16:9)	720 × 1280	reference layout
720 × 1560 (19.5:9, cutout phones)	720 × 1560	content spreads vertically; board centred; empty space above/below board on easy levels
540 × 960 (small phone)	720 × 1280 (scaled 0.75)	identical layout, smaller physical size
1200 × 1920 (tablet portrait)	800 × 1280	centred 780-px column with 10 px side margins
1280 × 720 (landscape/tablet/foldable)	2275 × 1280	centred 780-px phone-style column; large empty side areas

13. Status of the review findings (revision 3)

Resolved and verified (tests and re-rendered screens): 1. Contrast failures (primary button text, captions, switch labels, tagline, coral graphics) → §11, `tests/test_contrast.gd`. 2. Rule tags for reserved and one-way trays → badge on the tray's top edge (screens 21, 22), wording aligned in intro cards, rules text and refusal toasts. 3. Padlock/ice legibility → status pills that leave the shape visible (screens 20, 23, 24). 4. Toast duration scales with text length; Restart moved away from Hint; board position and menu spacing; staged reward animation. 5. **Amber triangle** (weakest object): fill `#F5B400 → #EAA600` and a 50 %-darker, thicker rim for that shape only; rim now 5.1:1 on the tray, fill 1.46:1, hue still yellow-orange and ≥ 50 RGB-distance from the orange cross; checked normal / selected / locked / frozen / stacked / on page background (`sheet_4_…`). No other shape was darkened. 6. **Tray boundary** → dedicated `TRAY_OUTLINE` token, 4 px, 3.59:1 on the page (was 2.65:1 at 3 px); tray fill unchanged to keep the paper look; verified on level 40. 7. **Hint state** → tests prove: the highlight persists until the next interaction (no timeout), Undo and Restart clear it and reset any pulsed object, a stale/illegal hint is never charged, insufficient coins explain accurately and charge nothing, double-pressing the confirmation charges once. 8. **Dialogs** → tests prove each dialog (pause, rules, hint, insufficient coins, restart, win, fail) has ≤ 1 coral primary, the destructive restart uses the danger style, and win/fail dialogs are not dismissible. 9. **Legibility on small layouts** → across all 40 levels the smallest slot is 80 logical px on every phone aspect (guarded by a test ≥ 76): rule tags ≈ 25 px radius, status pills ≈ 30 px tall with ≈ 25 px digits.

Evaluated, deliberately not changed (provisional until real-device use): * **Pause position (top-left):** it is an infrequent action, Android Back also opens Pause during play, and the frequent actions (Undo/Hint/Restart) are in the bottom thumb zone. Not relocated; revisit only if device testing or

playtests show reach problems. * **Level select:** current level (coral border), completed levels (gold stars; a cleared level always shows ≥ 1 gold star while an unplayed one shows none), locked levels (padlock) and stage headers already communicate state. No extra tick or stage counter was added to avoid clutter.

Still open: * Tray fill vs page background is 1.20:1 (the outline carries the boundary). * No haptics (deliberate: needs the `VIBRATE` permission). * TalkBack/screen reader impossible with Godot 4.7.1 (no Android accessibility server). * No persistent symbol legend (long-press help); hint flow is two taps plus a dialog. * Single default font with synthetic bold; landscape/tablet is a centred phone column. * **Audio, difficulty fairness and real-device touch/reach have never been tested by humans or on a device** (`docs/PLAYTEST_PLAN.md`, `docs/DEVICE_TEST_PLAN.md`: not run).

14. Ready-to-paste review request

> You are a senior mobile game UI/UX and visual designer. Attached: a design specification for a finished Android puzzle game (“Unjam: The Last Slot”), three contact sheets and 29 individual screens. Please (1) review the visual design, information hierarchy, interaction model, copy and accessibility against current mobile-game best practice and WCAG 2.2 / Material touch-target guidance; (2) challenge each open item in section 13 — agree, disagree or reprioritise; (3) list additional problems you see in the screenshots; (4) propose **specific, minimal, implementable changes** (hex values, sizes, copy, animation timings) ranked by impact vs. effort, marking which are *must-fix before launch*, *should-fix*, and *nice-to-have*; (5) say what you would test with real players. Constraints: portrait phone game, Godot 2D with procedurally drawn graphics (no image assets), offline, no ads, no custom font files, small team. Do not suggest adding new game mechanics or monetisation.

15. Appendix

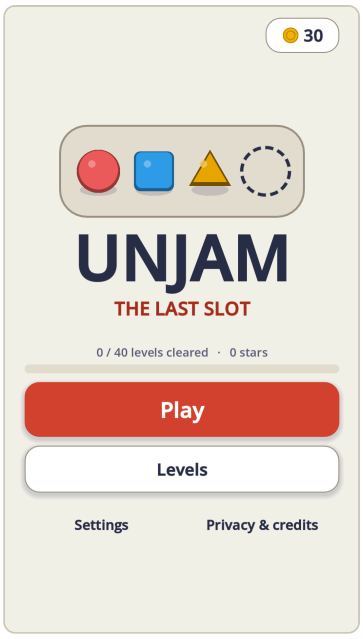
Where the UI lives (source): `src/ui/ui_theme.gd` (tokens + button/label factories) · `shapes.gd` (object shapes + icons) · `object_view.gd`, `tray_view.gd`, `board_view.gd` (board + animation) · `modal.gd`, `icon_button.gd`, `toggle_row.gd`, `settings_panel.gd`, `logo.gd`, `column_margin.gd` · screens: `splash_screen.gd`, `menu_screen.gd`, `level_select_screen.gd`, `game_screen.gd`, `credits_screen.gd` · navigation `main.gd` · text `src/core/campaign_text.gd`. **Regenerate the screenshots:** `xvfb-run -a godot --path . --rendering-driver opengl3 --resolution 1080x1920 --script res://tools/make_design_screens.gd` (outputs to `docs/design/screens/`). **Rules and economy:** `docs/RULES.md` · **Project status:** `SSOT.md`.

Appendix A — Current screens (all 29, revision 3)

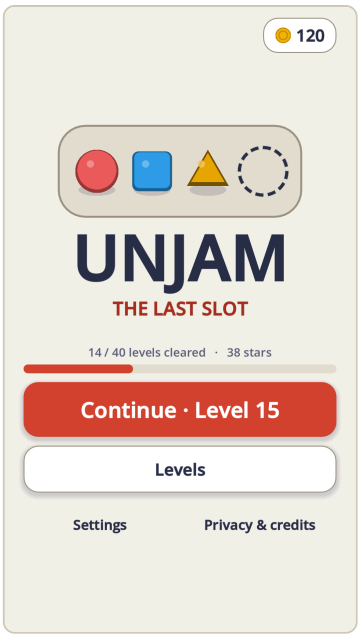
Each image is the real game rendered at 1080×1920 (shown reduced).



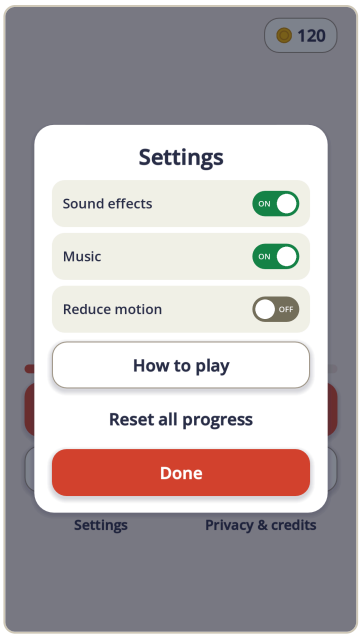
01_splash



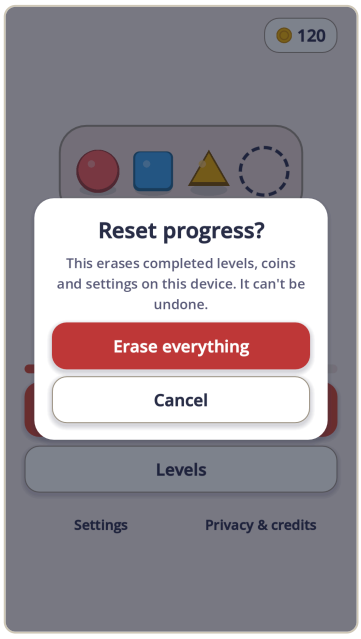
02_menu_new_player



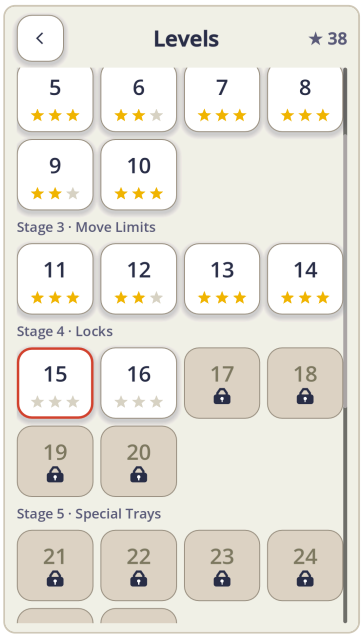
03_menu_with_progress



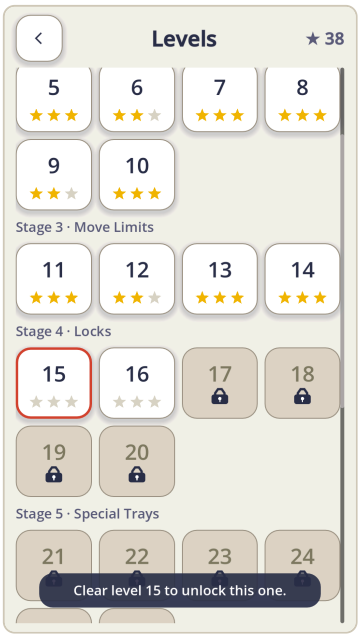
04_settings_dialog



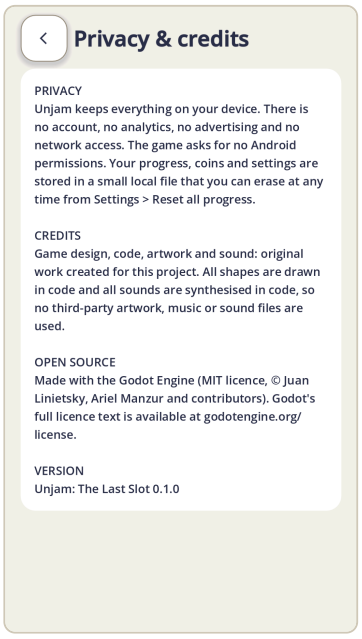
05_reset_progress_confirm



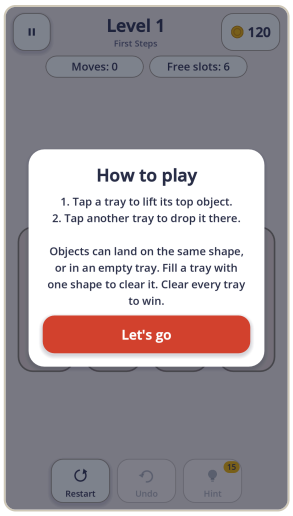
06_level_select_top



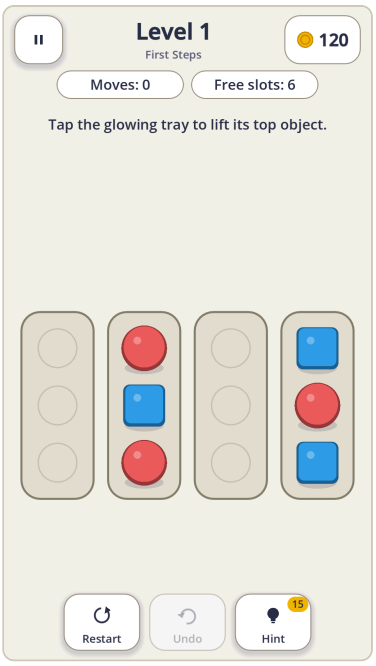
07_level_select_locked_toast



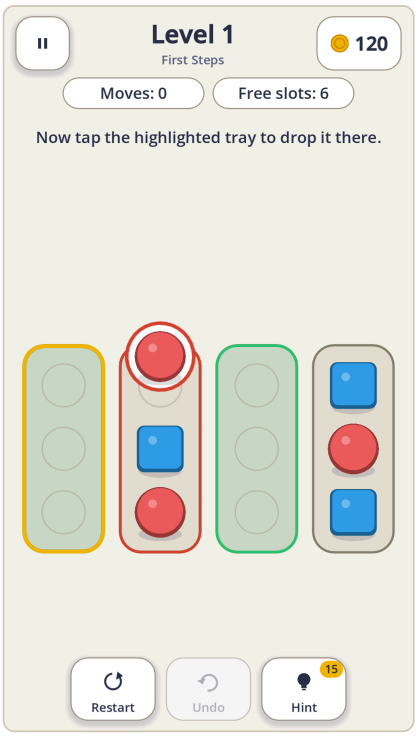
08_privacy_and_credits



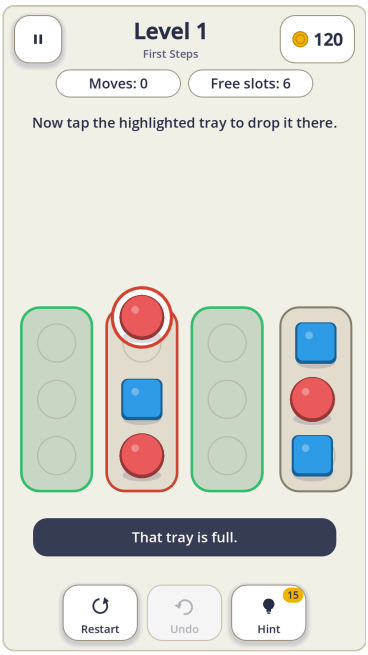
09_intro_card_how_to_play



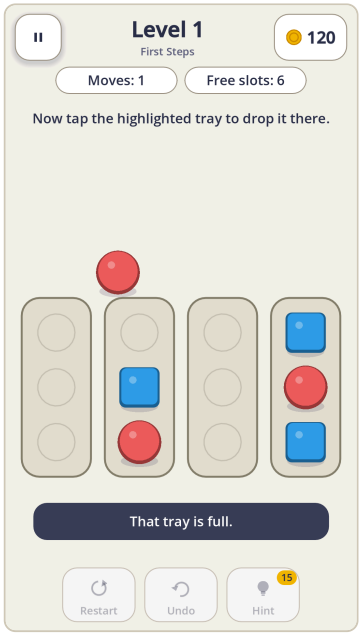
10_game_level1_initial_with_coach



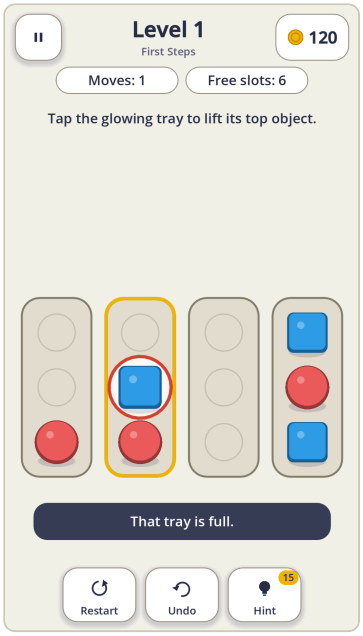
11_game_selected_object_and_targets



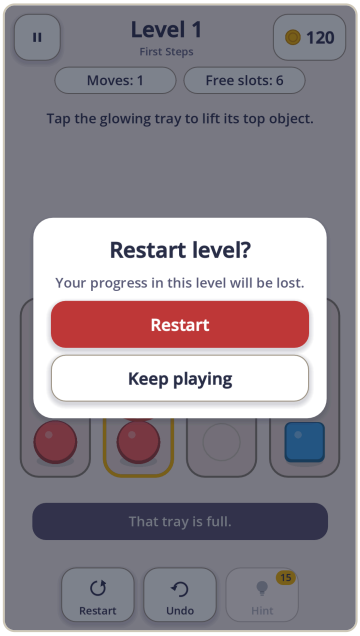
12_invalid_move_toast_and_shake



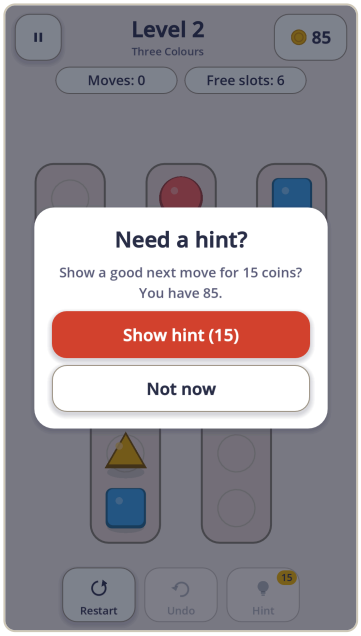
13_move_in_flight



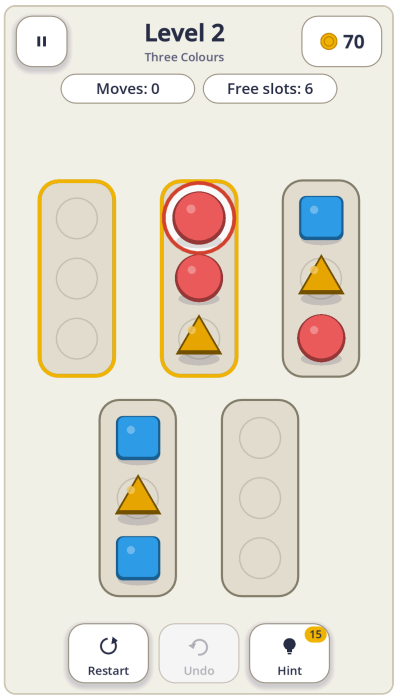
14_after_move_undo_enabled



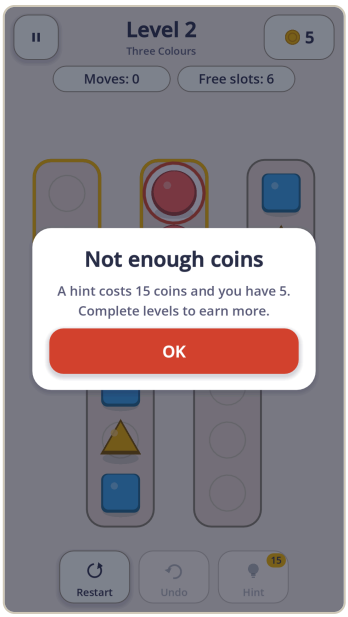
15_restart_confirmation



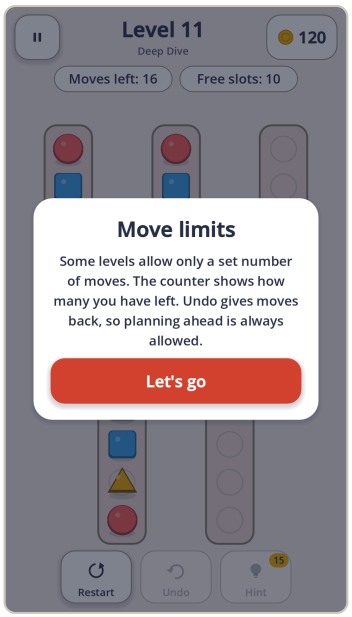
16_hint_confirm_dialog



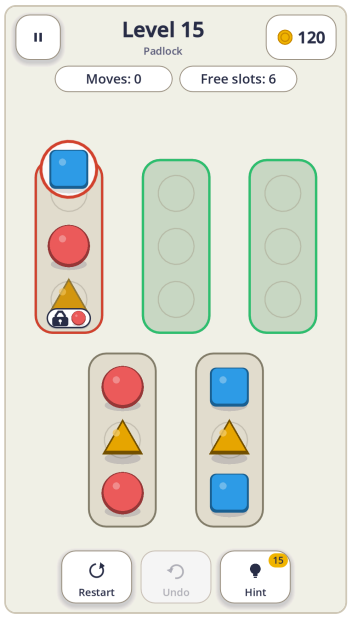
17_hint_highlight_source_and_target



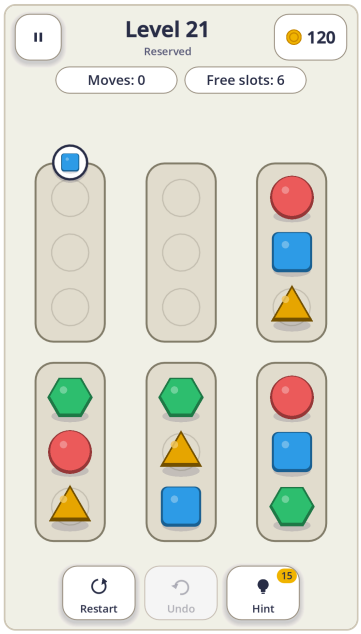
18_hint_not_enough_coins



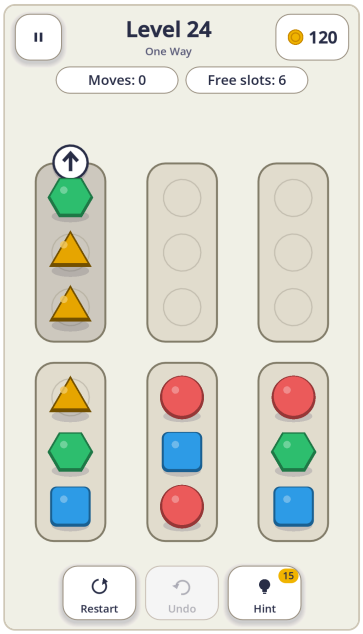
19_level11_move_limit_intro



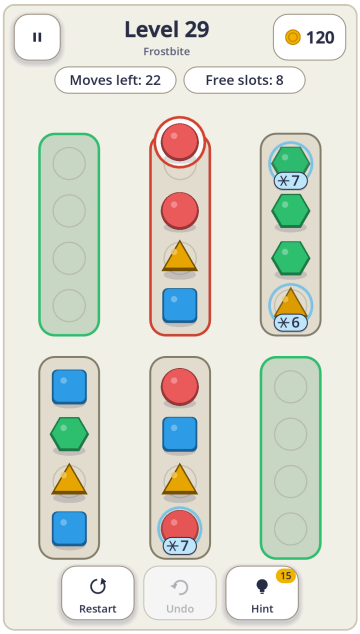
20_level15_padlocks



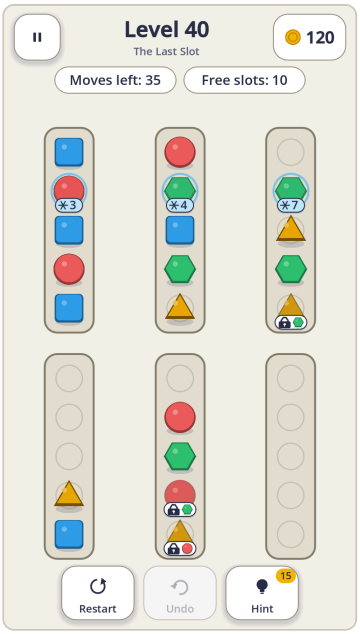
21_level21_reserved_trays



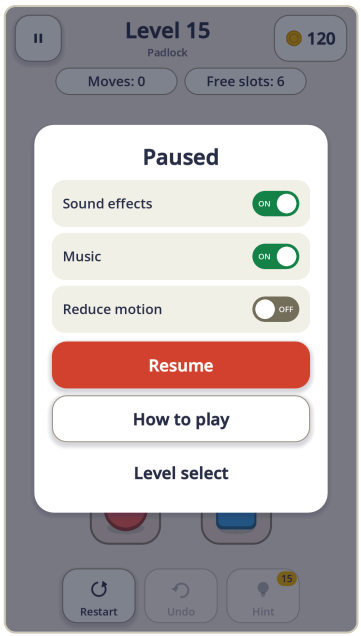
22_level24_one_way_tray



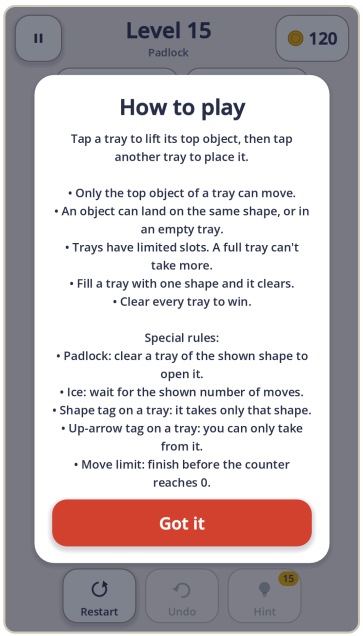
23_level29_ice_frozen_objects



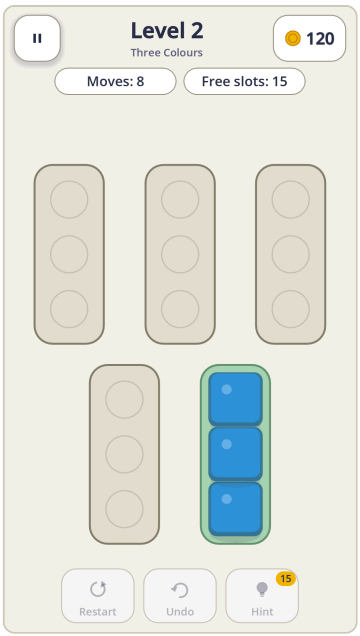
24_level40_dense_mastery



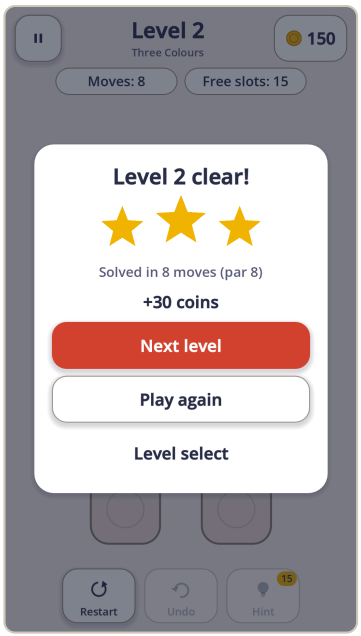
25_pause_menu



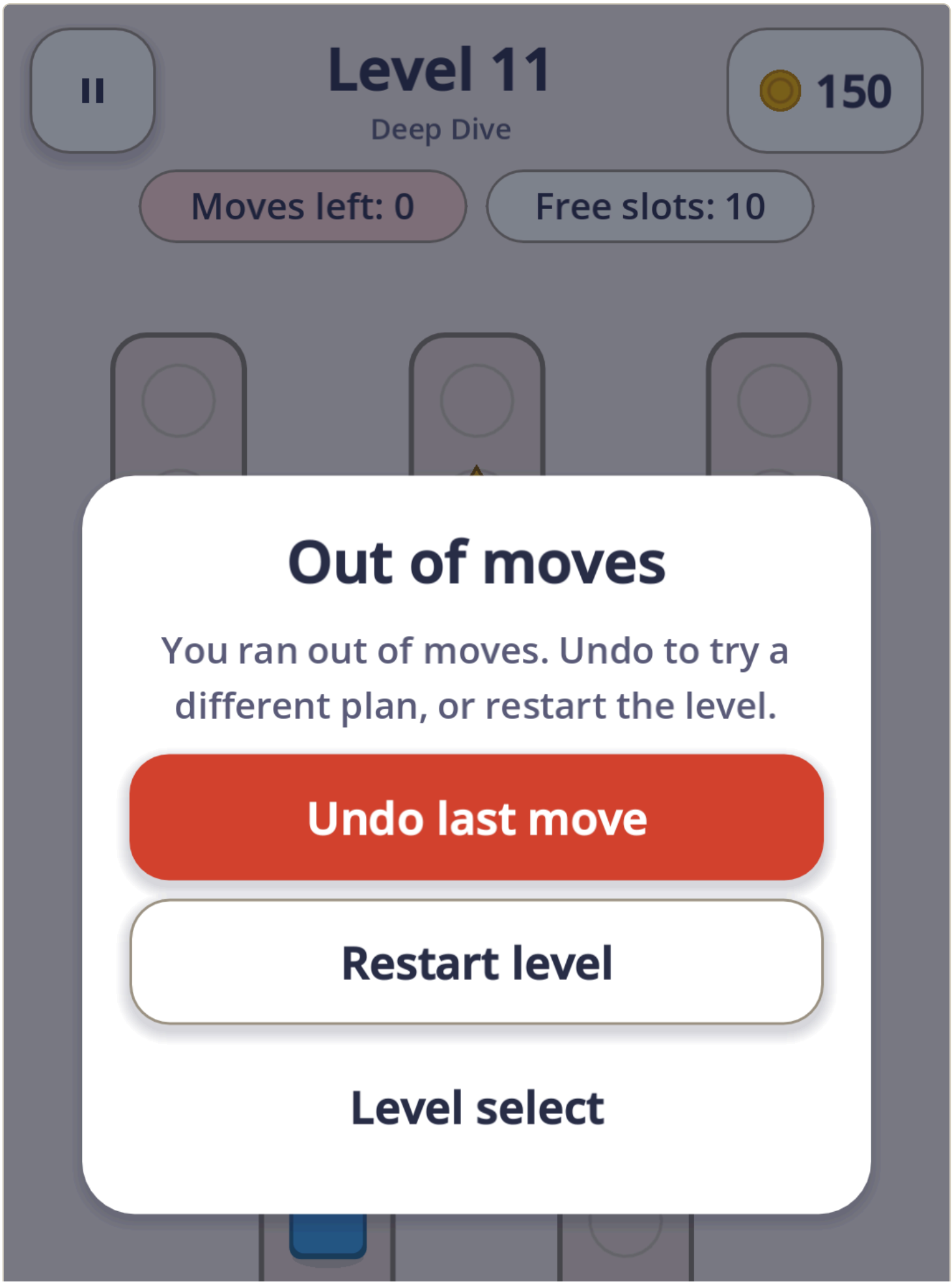
26_how_to_play_rules_dialog



27_clear_flash_before_win_dialog



28_level_complete_dialog


II
Level 11
Deep Dive
150
Moves left: 0
Free slots: 10
Out of moves
You ran out of moves. Undo to try a different plan, or restart the level.
Undo last move
Restart level
Level select



29_level_failed_dialog

Appendix B — Amber triangle and all shapes, before (top) and after (bottom)


normal
selected
locked (key: blue)
frozen 4
on page bg
normal
selected
locked (key: blue)
frozen 4
on page bg